



MECH MAINT- BOILER Department
SIMHADRI



Ref:SMTTP/MECH MAINT-/2022-23/LMI/366215

Date:

18/08/2022(09:40:52)

Subject:Review of LMI “ATTEMPERATOR SPRAY NOZZLES AND LINERS

OPERATION & MAINTENANCE

MECHANICAL MAINTENANCE

(PRESSURE PARTS)

REF.: SMPP/O&M/MM/PP/

Date: 18.08.2022

Subject: Review of LMI “ATTEMPERATOR SPRAY NOZZLES AND LINERS”

LMI No: LMI/OGN/OPS/MECH/002

The above approved LMI was prepared in the month of Feb.’2004. The LMI was revised in 2010, 2013,2020 and 2021. The envisaged changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

Submitted by:

राहुल जैन - वरिष्ठ प्रबंधक

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On:18/08/2022(09:40:52) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/366215

The subject LMI has been reviewed. The suggested changes in LMI as per COS-ISO-00-OGM/OPS/MECH/02 REV-01 Aug'22 have been carried out.

May please accord approval.

Submitted by:

सुरेश बाबु कोल्ला - उप महाप्रबंधक

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On:18/08/2022(12:22:16) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/366215

Following changes incorporated in revised LMI (attached as annexure -NEW LMI)

- a) Clause no. 6.0: Inspection procedure incorporated
- b) Clause no. 7.0: Figures of spray nozzles incorporated
- c) Annexure-I: Checklist for Attemperator inspection incorporated

May please be approved

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

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Submitted for approval pls

Submitted by:

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May please be accorded approval.

Submitted by:

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On:19/08/2022(14:53:45) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/366215

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

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On:19/08/2022(16:18:15) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/366215

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/002/

ISSUE NO.1; REVISION NO. 4.00; DATE: AUG' 2022

**ATTEMPERATOR SPRAY NOZZLES
AND LINERS**

APPROVED FOR

***Implémentation*-----**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	ATTEMPERATOR SPRAY NOZZLES AND LINERS
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/MECH/002/</u>
b) REVISION/ISSUE NO.&DATE	<u>ISSUE-1; REV.-4.00; AUG'2022</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>OGN/OPS/MECH/002</u>
<u>4.PREPARED BY</u> SENIOR MANAGER	SIGNATURE-----
<u>5. CHECKED BY</u> DGM (MM-PP) AGM (BMD) GM (MTC.) GM (O&M)	SIGNATURE----- SIGNATURE_____ SIGNATURE_____ SIGNATURE_____
<u>6. REVIEWED AND APPROVED BY</u>	SIGNATURE_____ HOP, SIMHADRI
<u>8.DATE OF ISSUE</u>	
<u>9.INDEX LIST</u>	ENCLOSED.
<u>10.DISTRIBUTION LIST</u>	ENCLOSED.

NTPC	LMI	SIMHADRI
TITLE: ATTEMPERATOR SPRAY NOZZLES AND LINERS		
REV: 4.00 DATE: AUG'2022		
REF NO. LMI/OGN/OPS/MECH/002/		PAGE : 1 OF 7
	<u>1.0 INDEX</u>	
<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
1	INDEX	1
2	INTRODUCTION	2
3	SCOPE	2
4	REFERENCE DOCUMENT	2
5	SUPERSEDED DOCUMENT	2
6	SPECIFIC INSTRUCTIONS	2
7	METHODOLOGY OF IMPLEMENTATION & RESPONSIBILITY CENTERS	3
8	EXCEPTION REPORTING	3
9	RESPONSIBILITY	3
10	RECORD KEEPING	4
11	REVIEW	4
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13	DISTRIBUTION LIST	6
14	COMPLIANCE RECORD	7

NTPC	LMI	SIMHADRI
TITLE: ATTEMPERATOR SPRAY NOZZLES AND LINERS		
REV: 4.00 DATE: AUG'2022		
REF NO. LMI/OGN/OPS/MECH/002/		PAGE : 2 OF 7
<u>2.0 INTRODUCTION</u> At Simhadri two spray type desuperheaters are installed in the connecting link between the superheater low temperature pendant outlet header and the superheater division panel inlet header for control of MS temperatures. Two spray type desuperheaters are installed in the reheater inlet leads near the reheater inlet header for control of reheater temperatures. Temperature reduction is accomplished by spraying water into the path of steam through a nozzle at the entering end of desuperheater. There have been reported cases of cracking and complete detachment of attemperator spray nozzles, due to fatigue strain which has resulted in blockage and overheating of downstream tube elements requiring extended outages to locate and remove the broken parts. Similarly, incidents of attemperator liner detachment and displacement have been reported resulting in blockage of steam outlet flow and temperature imbalance across the boiler width. The cracking of nozzle welds and liners is not detectable during routine plant monitoring. The objective of this LMI is to establish an inspection strategy for attemperator spray nozzle and liner so that informed decisions can be made with respect to maintenance to minimize forced outages. <u>3.0 SCOPE</u> The scope of this LMI is to detail the methods and periodicity that are to be adopted for the inspection of attemperator spray nozzles and liners. <u>4.0 REFERENCE DOCUMENT</u> Operation guidance note OGN/OPS/MECH/002 circulated by operation services - NTPC corporate center. <u>5.0 SUPERSEDED DOCUMENT</u> ATTEMPERATOR SPRAY NOZZLES AND LINERS REV: 2.00 DATE: APR'2013 <u>6.0 SPECIFIC INSTRUCTIONS</u> To achieve the objective of this LMI, the following approach is to be followed: - Periodic Inspection of attemperator spray nozzles and liners with videoscope. - Remedial Actions to sustain integrity of attachments on discovery of internal cracks through videoscope involving cutting of the coaxial tube. Taking and saving snap shots of the spray nozzle and liner and filling Annexure-1 with FQA along with the protocols in OH report files.		

NTPC	LMI	SIMHADRI	
TITLE: ATTEMPERATOR SPRAY NOZZLES AND LINERS			
REV: 4.00 DATE: AUG'2022			
REF NO. LMI/OGN/OPS/MECH/002/		PAGE : 3 OF 7	
7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS			
a) All the superheater and reheater attemperator nozzles should be inspected during overhauls once in six years using dye penetrant or MPI for the integrity of welds as per the attached SH and RH DESUPERHEATER drawing.			
FREQUENCY - once in six years RESPONSIBILITY - HOS (Pressure Parts)			
b) All the attemperator body liners should be inspected for structural rigidity and fixing attachment integrity during overhauls using endoscope, once in six years.			
FREQUENCY- once in five years RESPONSIBILITY- HOS (Pressure Parts)			
c) Strengthening of welds/ replacement of liners is to be carried out if any defects are observed during inspection as stated above.			
FREQUENCY- As and When necessary RESPONSIBILITY- HOS (Pressure Parts)			
8. EXCEPTION REPORTING			
The exceptions (cracks in nozzle welds/ liners) encountered during inspection are to be reported as per the following format to HOD (O&M) ----- HOS (Pressure Parts)			
Description of exception	Unit No.	Location of exception	Remarks
9.0 RESPONSIBILITIES			
The responsibility of carrying out different tasks in relation to inspection and maintenance of integrity of attemperator nozzles and liners as detailed in the LMI are with HOS (Pressure Parts).			

NTPC	LMI	SIMHADRI
TITLE: ATTEMPERATOR SPRAY NOZZLES AND LINERS		
REV: 4.00 DATE: AUG'2022		
REF NO. LMI/OGN/OPS/MECH/002/		PAGE : 4 OF 7
<u>10. RECORD KEEPING</u> The records to be maintained under this LMI include: a) Compliance record of the activities detailed under this LMI b) Reports of survey and rectification work carried out c) Exception reporting The above records are to be maintained by HOS (Pressure Parts) <u>11.REVIEW</u> This document shall be reviewed by HOP on TWO yearly basis.		

NTPC		LMI		SIMHADRI
TITLE: ATTEMPERATOR SPRAY NOZZLES AND LINERS				
REV: 4.00 DATE: AUG'2022				
REF NO. LMI/OGN/OPS/MECH/002/			PAGE : 5 OF 7	
12.0 REVISION INDEX (for changes/revision)				
Sl. No.	Rev. no.	Date	Details	Approved by
1	001	15.04.10	i) Approval routing changed at “LMI APPROVAL DOCUMENT”	
			ii) Frequency of inspection is reviewed from 5 to 6 years.	
			iii) Responsibility changed from SS (PP) to In charge (PP) at responsibilities	
			iv) Reviewing authority changed from DGM (MM) to HOP (I/C)	
			v) Distribution List	
2	002	30.04.13	i) Responsibilities changed from In Charge (MMD-PP) to HOS (PP) at Responsibilities.	
			ii) Inclusion of Unit#3&4 in compliance record	
			iii) Distribution List	
3	003	13.01.2020	i) Approval routing changed at “LMI APPROVAL DOCUMENT”	
			ii) Review Frequency changed from "Three years" to "TWO" years	
			iii) Distribution List	
4	003	02.12.2021	No changes done.	
5	004	17.08.2022	a)Clause no. 6.0: Inspection procedure incorporated b)Clause no. 7.0: Figures of spray nozzles attached. c)Annexure-I: Checklist for Attemperator inspection incorporated	

[illegible]

NTPC	14. COMPLIANCE RECORD	SIMHADRI
TITLE: ATTEMPERATOR SPRAY NOZZLES AND LINERS		
REV: 4.00 DATE: AUG'2022		
REF NO. LMI/OGN/OPS/MECH/002/		PAGE : 7 OF 7
UNIT NO.	ACTIVITY	COMPLIANCE
I/II/III/IV		

ANNEXURE-I

1) Checklist/ Protocol for Attemperator Inspection

Station & Unit No.:

Attemperator SH/ RH:

Attemperator Location:

Attemperator Type (Probe type/ Circumferential type):

Running hour of Unit: Hrs.

History of Inspection:

History of Failure and its repair:

Stage wise LHS and RHS SH Spray flow (Average/ Maximum Flow in T/ Hrs.):

S.No.	Description	Inspection during every overhauling	Inspection during Short Shutdown/ Operation	Yes/ No	Detail of Exceptions
1	Setpoint for Desuperheated steam temperature above saturation temperature downstream of each attemperator set at 20°C	YES			
2	Desuperheated steam downstream of each attemperator in SH / RH is maintaining 20°C above saturation temperature	YES			
3	Attemperator spray nozzle checked for partial/ full chokage during operation		Yes		
4	Material Grade checking of SH Spray pipe, elbows, Attemperator by XRF alloy analyser	YES			
5	DPT/ MPI of all weld joints (Butt weld, Fillet welds) in pipe downstream of Spray CV, pipe upstream and downstream of attemperator near attemperator.	YES			
6	Inspection of support and hangers of attemperator for excessive pipe movement/ damaged support/ topped out or bottom out hangers in hot or cold condition etc.	YES			
7	Inspection of attemperator and its downstream pipe for sagging/ bowing	YES			
8	Any water droplets coming out from insulation of spray pipe during attemperator in service		Yes		
9	Visual inspection of Upstream and Downstream pipe & elbows near to attemperator and attemperator for crack from outside	YES			
10	Thickness Survey of downstream pipe, elbows of spray CV from outside for erosion.	YES			

12	Thickness Survey of upstream and downstream pipe, elbows of attemperator and near attemperator from outside.	YES			
11	If material of Spray piping is carbon steel, the FAC to be checked by carrying thickness survey at bends, elbow, reducers, valves etc.	YES			
13	Videoscope of attemperator lining from first elbow upstream and downstream of the attemperator and to be checked for following:	YES			
	a) Condition of nozzle for crack and condition including size of its atomising hole	YES			
	b) Inspection of support and position screws/ weld of liners and rubbing of positioners with liner	YES			
	c) Inspection of liners, liner welds for crack and erosion	YES			
	d) Inspection of liners for shifting of its position axially	YES			
	e) Inspection of liners for its shifting to an eccentric position	YES			
	f) In case of crack in liner, fractured liner, all fractured components are checked and retrieved	YES			
	g) Visual inspection of Upstream and Downstream pipe and elbows of attemperator liner for crack or erosion from inside	YES			
14	SH Spray CV passing problem		YES		
15	SH Spray block valve passing problem		YES		
16	RH Spray CV passing problem		YES		
17	RH Spray block valve passing problem		YES		

BMD

OPERATION

FQA

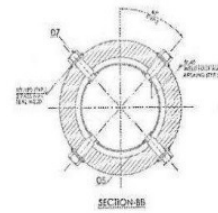
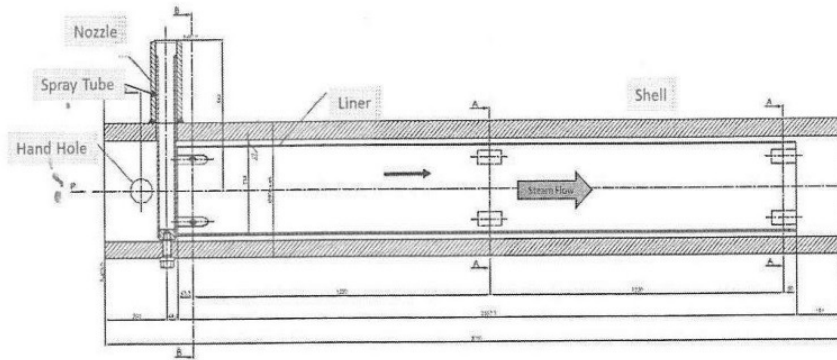
2) PROTOCOL FOR SH & RH SPRAY NOZZLE & LINER INSPECTION

STATION: _____

UNIT _____

Date -		Unit -	
Sl No	Description	Handhole cap (Dimension and Materials)	Inspection Remark
1	SH Spray Nozzle and Liner LHS		
2	SH Spray Nozzle and Liner RHS		
3	RH Spray Nozzle and Liner LHS		
4	RH Spray Nozzle and Liner RHS		
BMD		FQA	

SH-DESUPERHEATER



RH-DESUPERHEATER

